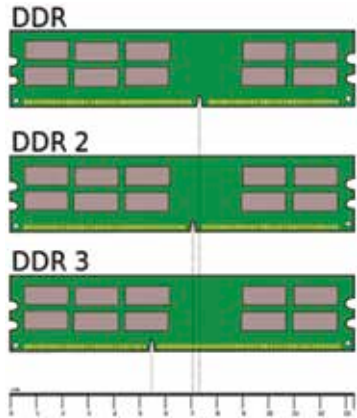


of bits that are transferred during one clock cycle. SDRAM transfers 8 bytes per clock cycle so when a PC133 RAM is considered it should transfer at $133 \text{ MHz} \times 8 \text{ bytes} = 800 \text{ MB/s}$. PC266 transfers at 2100 MB/s . This was when memory worked in a single channel mode, current memory works in Dual channel, i.e. 2 simultaneous connections. So a DDR2 800MHz RAM transfers data at 12800 MB/s instead of 6400 MB/s . This is a pretty huge gap and is the reason for a major performance boost when the dual channel memory kits started coming out.



2. Timing - Memory is like a grid made out of rows and columns. Since this form of memory is a random access memory the way a particular bit of data is accessed by referring to a row and the column of the memory. The signal is aptly named CAS (Column Access Strobe) and the delay that exists after the CAS signals a particular memory cell is called CAS Latency or CL. This time delay is based upon the number of clock cycles and are not milli-seconds as is commonly mistaken. The typical timing notation is a combination of 4 numbers - tRCD, tRP, tRAS and CL. They look like this - 7-7-7-21 or 9-9-9-24. What we will be focussing on is the CL timing which is the last number in the sequence. Lesser CL results in faster fetching of data from the memory module and they are equally expensive as well.

3. Capacity - A common concern when figuring out what RAM is to get how much you'll need. This depends on the type of applications being used, simple ones like spreadsheet programs will utilise a few MBs while 3D model rendering will use as much RAM as is available. Games use the VRAM that is available on GPUs for storing texture data and computer RAM is used to store level data. A simple method is to double whatever you need. If your OS and a few programs run fine on 4 GB for RAM then get 8 GB instead. If there is less RAM available then the cache is reduced or emptied on a regular basis. So applications load slower since all applications vie for whatever little memory is available. As a rough approximation, low end